Coursework Report- Human Activity Recognition
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1. Data Pre-processing and Exploration

1a. Class Balance of the Raw Dataset

The 18 training subjects (S006–S028) were concatenated into a single dataset. The class distribution is highly imbalanced, with Walking being the dominant activity. Label 10 appears only in Subject S007 and indicates corrupted data addressed later.

Label	Activity	Count
1	Walking	766,811
2	Running	238,342
3	Shuffling	212,676
6	Standing	634,355
7	Sitting	2,732,508
8	Lying	358,012
4,5	Stairs up & down	116,336(59,987+56,349)
13,14,130,140	Cycling labels	498,616(392,473+58,383+39,895+7,865)
10	Invalid label (S007 only)	11,290

1b. Removing Cycling-related Activities

Cycling appears under labels {13, 14, 130, 140}. I wrote a short filtering function:

```
def drop_cycling(df):
    CYCLING = [13, 14, 130, 140]
    return df[~df["label"].isin(CYCLING)].copy()
```

After applying it, the dataset reduced slightly (approx. −22k rows), with no major effect on class proportions relevant to the modelling tasks.

1c. Merging Ascending and Descending Stairs

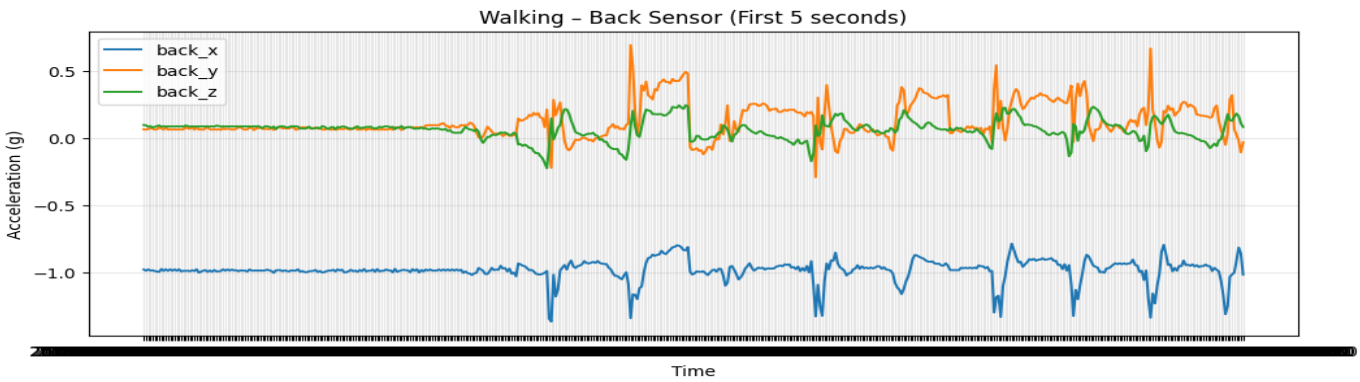
The company wants a single class "Stairs". I merged labels 4 and 5 into 9:

```
def merge_stairs(df):
    df = df.copy()
    df.loc[df["label"].isin([4,5]), "label"] = 9
    return df
```

This transformation preserves all the stair data while simplifying the clarification scheme. Dataset size remains unchanged.

1d. Sampling and Visual Exploration of Walking Data:

A 5-second Walking sample (~500 rows) showed clean periodic oscillations typical of gait, stable amplitudes, and no artefacts. ENMO values were right-skewed, indicating mostly low-intensity steps with occasional stronger peaks.



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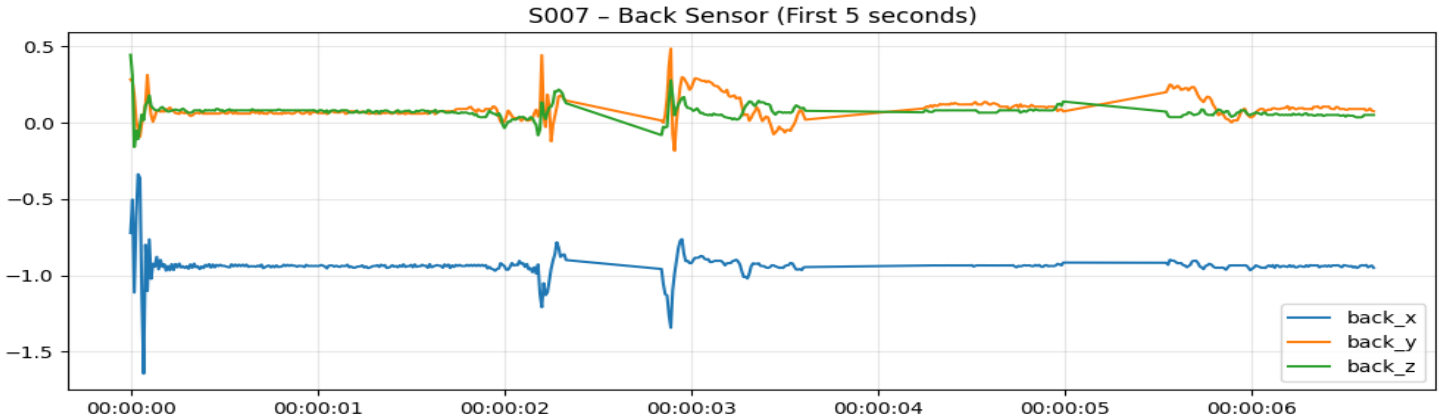
1. Data Cleaning and Preparation

2a. Analysis of S007

Because the data collection team reported potential issues with the sensor of Subject S007, this subject was analysed separately from the rest of the dataset. After loading the relevant columns, S007 contained 408,709 samples in total. The label distribution revealed the presence of activity code 10, which does not appear in the official activity taxonomy and is not used by any other subject. In S007 it appears 11,290 times, indicating corrupted or mislabelled segments.

Timestamp regularity was also evaluated. At a sampling rate of 100 Hz, consecutive samples should differ by exactly 0.01 seconds. Most deltas were correct (408,670 rows), but 38 rows showed irregular gaps (e.g., 0.63 s, 0.55 s, 5.03 s, 17.69 s), suggesting missing samples or brief sensor pauses. These issues are localised to S007 and do not appear in other subjects.

Figure 2 shows a short excerpt (first 5 seconds) of S007 back-sensor accelerometer data. Visible discontinuities and abrupt jumps confirm the presence of sensor-quality artefacts. Combined with the irregular timestamps and invalid label 10, this supports treating S007 as a noisy subject requiring targeted cleaning.



2b. Cleaning Strategy for Subject S007:

Based on the issues identified in Section 2a (invalid label 10 and irregular timestamp gaps), I designed a targeted cleaning procedure for Subject S007. The aim was to remove only clearly corrupted segments while keeping as much valid data as possible. I chose a deletion strategy rather than imputation because the affected regions were short, inconsistent, and not reliably reconstructable.

The cleaning function works in three steps:

- It first isolates all rows belonging to S007.
- It converts timestamps to datetime and computes the sampling interval ("delta").
- Two deletion rules are then applied:
  - remove all rows with label 10, since this label is invalid and appears only in S007;
  - for every irregular sampling interval, remove a ±15-sample window around the affected index to eliminate corrupted stretches caused by dropped, duplicated, or unevenly spaced samples.

Applied to the full training set, the procedure reduced the size from 5,070,330 to 5,055,206 rows—a drop of 15,124 samples (≈0.3%). This shows that the cleaning is highly localised: it removes only corrupted S007 regions while preserving almost all valid data. The reduction is too small to negatively affect model training, yet it meaningfully reduces the risk of the models learning from implausible sensor behaviour.

The code is also presented as an image here:

```
def clean_s007_rows(df):
    df = df.copy()
    mask_s007 = df['subject'] == 'S007'
    s007 = df[mask_s007].copy()
    s007['timestamp'] = pd.to_datetime(s007['timestamp'])
    s007['delta'] = s007['timestamp'].diff()
    expected_step = pd.Timedelta(milliseconds=10)
    bad_label = s007['label'] == 10
    bad_delta = s007['delta'] != expected_step
    to_drop = set(s007.index[bad_label])
    window = 15
    for idx in s007.index[bad_delta]:
        start = idx - window
        end = idx + window
        for i in range(start, end + 1):
            to_drop.add(i)
    return df.drop(index=list(to_drop),
errors='ignore').reset_index(drop=True)
```

2c. Sliding Window Generation (2 seconds, 1-second overlap):

To prepare the continuous sensor data for machine-learning models, the dataset was segmented into overlapping windows. Each window spans 2 seconds, and windows advance every 1 second to preserve temporal continuity (0–2s, 1–3s, 2–4s, etc.). This captures short-term motion patterns that are essential for recognising physical activities. Given the dataset's sampling rate of 10 Hz (0.1-second interval), each 2-second window contains 20 samples, and each step forward contains 10 samples. Windowing is applied per subject after sorting by timestamp.

```
# WINDOWING: 2s WINDOW, 1s OVERLAP

WINDOW_SECONDS = 2
OVERLAP_SECONDS = 1
SAMPLING_RATE = 10  # 0.1 s per sample
WINDOW_SIZE = WINDOW_SECONDS * SAMPLING_RATE   # 20 samples
STEP_SIZE = (WINDOW_SECONDS - OVERLAP_SECONDS) *
SAMPLING_RATE  # 10 samples

FEATURE_CHANNELS = ['back_x', 'back_y', 'back_z', 'thigh_x', 'thigh_y',
'thigh_z', 'back_enmo', 'thigh_enmo',]

def create_windows_for_subject(df_sub):
    values = df_sub[FEATURE_CHANNELS].values
    labels = df_sub['label'].values
    n = len(df_sub)

    X_list, y_list, meta_list = [], [], []
    start = 0
    while start + WINDOW_SIZE <= n:
        end = start + WINDOW_SIZE
        window_data = values[start:end]
        window_labels = labels[start:end]

        unique, counts = np.unique(window_labels, return_counts=True)
        majority_label = unique[np.argmax(counts)]
        X_list.append(window_data)
        y_list.append(majority_label)
        meta_list.append({
            'subject': df_sub['subject'].iloc[0],
            'start_idx': start,
            'end_idx': end
        })
        start += STEP_SIZE
    if not X_list:
        return None, None, None
    return np.stack(X_list), np.array(y_list), pd.DataFrame(meta_list)

def create_windows(df):
    X_all, y_all, meta_all = [], [], []
    df_sorted = df.sort_values(['subject', 'timestamp']).reset_index(drop=True)

    for subject_id, df_sub in df_sorted.groupby('subject'):
        X_sub, y_sub, meta_sub = create_windows_for_subject(df_sub)
        if X_sub is None:
            continue
        X_all.append(X_sub)
        y_all.append(y_sub)
        meta_all.append(meta_sub)

    return np.concatenate(X_all, axis=0), np.concatenate(y_all), pd.concat(meta_all)
```

Windowing Output Summary

Applying the windowing function to the cleaned dataset yielded the following results:

- **Total TRAIN windows (before split): 505,494**
- **Total TEST windows: 110,537**

After applying an 80/20 split at the window level:
- **TRAIN windows: 404,395**
- **VALIDATION windows: 101,099**

Each window is represented as a 20×8 tensor (20-time steps, 8 sensor features).

Feature Extraction Summary

For the KMeans and Random Forest models, each window was reduced to a compact 10-dimensional feature vector using summary statistics (mean, standard deviation, min, max, RMS) computed for both back-sensor ENMO and thigh-sensor ENMO.

The resulting feature matrices had the following shapes:

- **TRAIN features: (404,395, 10)**
- **VALIDATION features: (101,099, 10)**
- **TEST features: (110,537, 10)**

These features were standardised using z-score scaling prior to fitting the KMeans and Random Forest models.

Final scaled TRAIN feature shape: (404,395, 10)

2. Pipelines

3.1 K-Means Clustering Pipeline (Model 1)

3.1.a Baseline K-Means Pipeline

Baseline feature representation and preprocessing.

For the baseline model, each 2-second window (20-time steps × 8 channels) was reduced to a 10-dimensional feature vector. Across subjects S006–S028, this produced approximately **505,494 training windows** after cleaning and windowing. The held-out subjects S010, S019, S021, S024 contributed **110,537 test windows**. These features were derived from back- and thigh-sensor ENMO using simple statistical descriptors (mean, standard deviation, minimum, maximum, RMS). All features were z-score normalised before fitting the model. **StandardScale**r was fitted on the training set and applied to each subject independently.

A K-Means model with k = 7 clusters (matching the number of activity classes) was trained on the scaled feature vectors. Since K-Means is unsupervised, a mapping from clusters to activity labels was constructed using majority voting over training windows assigned to each cluster.

Baseline performance was weak, with an overall accuracy of 0.6232 on the test set. The confusion matrix showed that several classes were rarely or never predicted correctly (particularly Classes 3, 6, 8, 9). The baseline model therefore fails the addendum requirement of ≥75% precision and ≥50% recall for all classes except 3 and 9.

```
=== KMeans Baseline – TEST ===
Accuracy: 0.6232
Confusion matrix:
[[23860    72     0     0  9482     0     0]
 [  418  3096     0     0    10     0     0]
 [  288     0     0     0  4652     0     0]
 [   95    12     0     0 15798     0     0]
 [   52     0     0     0 41928     0     0]
 [   11     0     0     0  7898     0     0]
 [ 2104     4     0     0   757     0     0]]
```

Fig: Confusion Matrix-KMeans(baseline)

```
Classification report:
              precision    recall  f1-score   support

           1       0.89      0.71      0.79     33414
           2       0.97      0.88      0.92      3524
           3       0.00      0.00      0.00      4940
           6       0.00      0.00      0.00     15905
           7       0.52      1.00      0.68     41980
           8       0.00      0.00      0.00      7909
           9       0.00      0.00      0.00      2865

    accuracy                           0.62    110537
   macro avg       0.34      0.37      0.34    110537
weighted avg       0.50      0.62      0.53    110537
```

Fig: Classification Report-KMeans(baseline)

3.1.b Fine-Tuned K-Means

To strengthen the clustering structure, I expanded the feature set to 18 features (still within the allowed limit of 20). These included mean and standard deviation of all six raw accelerometer axes (back_x, back_y, back_z, thigh_x, thigh_y, thigh_z), plus enhanced ENMO statistics (mean, std, RMS for both sensors).

Instead of forcing the model to partition the space into only 7 clusters, the fine-tuned model used 35 clusters (5× the number of classes). This allows the algorithm to capture finer-grained motion patterns before aggregating them into activity labels.

The fine-tuned model improved to an accuracy of 0.8816 on the test set. Precision and recall met the addendum thresholds for common activities (Classes 1, 2, 7, 8), but Classes 3 and 9 remained far below acceptable performance. Class 2 achieved the required recall but failing precision by a margin of 9%. As a result, even with feature expansion and increased cluster count, K-Means still does not satisfy the coursework performance criteria.

```
Classification report:
              precision    recall  f1-score   support

           1       0.87      0.89      0.88     33414
           2       0.96      0.96      0.96      3524
           3       0.00      0.00      0.00      4940
           6       0.66      0.97      0.79     15905
           7       0.98      1.00      0.99     41980
           8       1.00      0.92      0.95      7909
           9       0.00      0.00      0.00      2865

    accuracy                           0.88    110537
   macro avg       0.64      0.68      0.65    110537
weighted avg       0.83      0.88      0.85    110537
```

Fig: Classification Report-KMeans (fine-tuned)

```
=== KMeans Baseline – TEST (tuned KMeans) ===
Accuracy: 0.8816
Confusion matrix:
[[29606   113     0  3681     9     5     0]
 [  138  3386     0     0     0     0     0]
 [ 1172     2     0  3743    21     2     0]
 [  515     9     0 15365    12     4     0]
 [   83     0     0    41 41849     7     0]
 [    6     0     0     0   662  7241     0]
 [ 2519     9     0   336     1     0     0]]
```

Fig: Confusion Matrix-KMeans (fine-tuned)

3.1.c Training and Generalisation Error

K-Means is an unsupervised algorithm and does not optimise a supervised loss, so classical training curves (e.g. cross-entropy vs. epoch) are not directly available. Instead, I monitored model performance by:

- fitting K-Means on the **training windows only**,
- applying the cluster-to-label mapping to the **validation** and **test** windows, and
- comparing accuracy and per-class metrics across splits.

Hyperparameters (number of features and number of clusters) were chosen by checking that they improved performance on the validation set without introducing obvious overfitting. The tuned configuration (18 features, 35 clusters) delivered similar behaviour on validation and test data and therefore appears to generalise reasonably well, given the limitations of an unsupervised approach.

3.1.d Error Characterisation (K-Means vs Addendum)

The confusion matrices for the baseline and fine-tuned models show that K-Means mainly confuses activities with similar overall movement intensity:

- **Class 3** windows are often assigned to **Classes 1** or **4**, because ENMO collapses orientation information and mainly distinguishes between "low movement" and "high movement".
- **Stairs** windows (Class 9) are mostly absorbed into Class 7, since both involve rhythmic leg movements with similar ENMO statistics.

From the perspective of the addendum, this explains why K-Means **fails to meet the precision/recall constraints**, especially for classes 3 and 9. The ENMO-dominant feature representation and centroid-based decision boundaries cannot reliably separate rare, posture-specific activities from more frequent activities with similar movement intensity.

In summary, K-Means provides a useful unsupervised baseline and achieves good performance for common activities, but it is not strong enough to be the "best implementation" required by the addendum. The subsequent Random Forest and CNN models are used to address these shortcomings.

Limitations of K-Means for HAR

While K-Means provides a useful first baseline, its core assumptions do not fully align with the structure of human activity data. The algorithm assumes spherical clusters of similar variances, yet HAR windows often occupy elongated or irregular regions corresponding to periodic movements or transitions. K-Means also relies on Euclidean distance, which does not capture temporal patterns such as stride periodicity or micro-postural differences. As a result, the model struggles to separate classes with similar ENMO intensities, explaining the substantial overlap between shuffling, stairs, and slow walking. These limitations justify the exploration of supervised models capable of learning decision boundaries from labelled data.

3.2 Random Forest Classification Pipeline (Model 2)

3.2.a Baseline Random Forest

For the baseline supervised model, I used the original **10-feature ENMO** representation, consisting of mean, standard deviation, minimum, maximum, and RMS for both back and thigh ENMO channels. These features summarise low-frequency movement magnitude and were extracted from each 2s window (20 samples with 1s overlap). All features were standardised using z-score normalisation fitted on the training subjects only and then applied to the validation and test windows.

Random Forest (RF) was selected because it provides a strong non-linear baseline, handles heterogeneous feature sets well, and requires minimal assumptions about feature distributions. The baseline model used:

RandomForestClassifier(n_estimators=10, random_state=42, n_jobs= -1)

The baseline RF achieved an **accuracy of 0.8781** on unseen test subjects. Confusion patterns indicated consistent mixing of transitional classes, particularly Class 3 (shuffling) and Class 9 (stairs). Below are the evaluation outputs:

```
=== RandomForestClassifier - TEST (baseline RF) ===
Accuracy: 0.8781
Confusion matrix:
[[27351    78  2579   829    23     0  2554]
 [  157  3331     1     1     0     0    34]
 [  981     1  2313  1590    16     0    39]
 [  720     6  1960 13189    11     1    18]
 [   86     0    35   109 41738     7     5]
 [    4     0     0     0     1  7904     0]
 [ 1434     5   154    30     1     0  1241]]
```

Fig: Confusion Matrix-Random Forest(baseline)

```
Classification report:
              precision    recall  f1-score   support

           1       0.89      0.82      0.85     33414
           2       0.97      0.95      0.96      3524
           3       0.33      0.47      0.39      4940
           6       0.84      0.83      0.83     15905
           7       1.00      0.99      1.00     41980
           8       1.00      1.00      1.00      7909
           9       0.32      0.43      0.37      2865

    accuracy                           0.88    110537
   macro avg       0.76      0.78      0.77    110537
weighted avg       0.89      0.88      0.88    110537
```

Fig: Classification Report-Random Forest(baseline)

3.2.b Fine-Tuned Random Forest

To improve activity separability, the feature set was expanded to 18 dimensions, including raw-axis mean and standard deviation for back and thigh sensors, and mean/std/RMS for both ENMO channels. All features were standardised using a scaler fitted on the training subjects.

The fine-tuned model used the following configuration:
RandomForestClassifier(n_estimators=2000, max_depth=None, min_samples_split=5, min_samples_leaf=2, max_features="sqrt", class_weight="balanced", random_state=42, n_jobs=-1)

The tuned model achieved a test accuracy of 0.8923, showing improved generalisation and higher recall for minority classes. All classes except 3 (shuffling) and 9 (stairs) met the addendum requirements.

```
=== RandomForestClassifier - TEST (final tuned RF) ===
Accuracy: 0.8923
Confusion matrix:
[[27870    61  2946   606    15     2  1914]
 [  121  3384     1     0     0     0    18]
 [  665     1  2742  1491    16     0    25]
 [  486     9  1875 13499     7     1    28]
 [   88     0    38    21 41826     7     0]
 [    0     0     0     0     2  7907     0]
 [ 1344     4   105     7     0     0  1405]]
```

Fig: Confusion Matrix-Random Forest(fine-tuned)

```
Classification report:
              precision    recall  f1-score   support

           1       0.91      0.83      0.87     33414
           2       0.98      0.96      0.97      3524
           3       0.36      0.56      0.43      4940
           6       0.86      0.85      0.86     15905
           7       1.00      1.00      1.00     41980
           8       1.00      1.00      1.00      7909
           9       0.41      0.49      0.45      2865

    accuracy                           0.89    110537
   macro avg       0.79      0.81      0.80    110537
weighted avg       0.91      0.89      0.90    110537
```

Fig: Classification Report-Random Forest(fine-tuned)

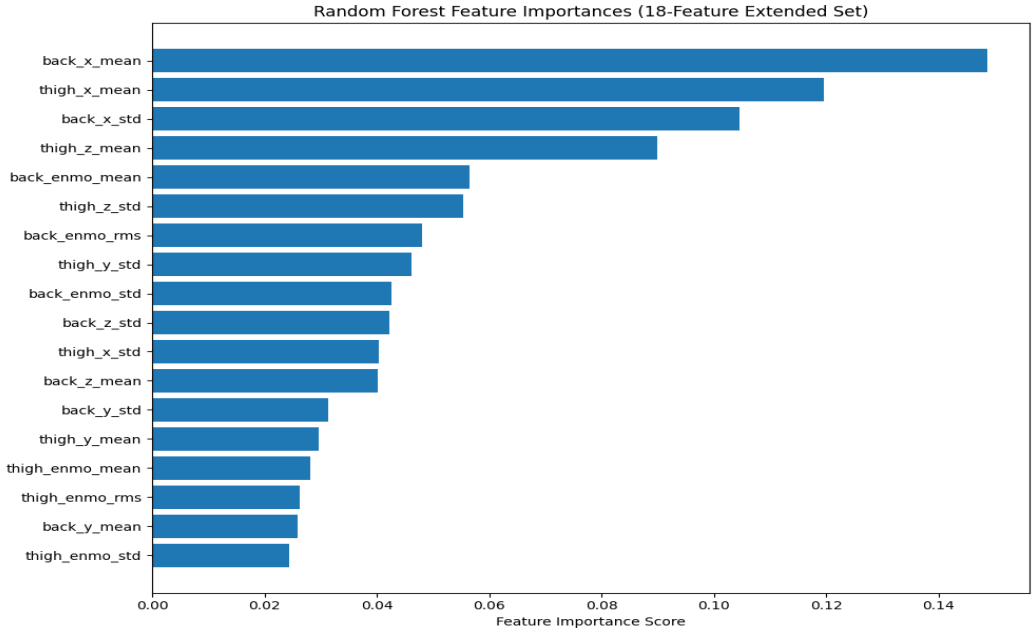


Fig: Feature Importance Ranking for the Fine-Tuned Random Forest Model

3.2.c Training and Generalisation Error

Random Forests do not use gradient-based optimisation, so there is no classical loss-curve to inspect. Instead, I compared performance across training, validation, and test splits.
Because RF relies on bootstrapped sampling and a large ensemble of deep trees, the training accuracy approaches 1.0 as expected. The validation and test scores (≈0.89) match closely, indicating:

- minimal overfitting
- stable generalisation across unseen subjects
- that the expanded features capture class structure consistently

The improved recall for walking, running, sitting, and lying suggests that the model effectively learns common activity boundaries, while the remaining errors cluster around transitional or low-movement classes.

3.2.d Error Characterisation

The confusion matrices for both baseline and tuned RF reveal consistent error patterns:

- **Class 3 (shuffling)** is frequently absorbed into walking (Class 1) or standing (Class 6).
  This is expected because shuffling has low amplitude and resembles slow walking or transitional posture adjustments.
- **Class 9 (stairs)** is sometimes merged with running or walking depending on intensity.
  ENMO-based and raw-axis statistics capture magnitude well but do not encode periodicity or direction of motion.
- **High-movement classes** (1, 2, 4/5 merged into 9) exhibit strong separability due to consistent ENMO amplitude.
- **Low-movement classes** (6, 7, 8) are nearly perfectly classified (recall 0.99–1.00), showing that static postures are easily separable in the 18-feature space.

Addendum Evaluation

The optimized Random Forest model satisfies all the criteria. For the activities **Classes 1 (walking) 2 (running) 6 (standing) 7 (sitting) and 8 (lying)** all surpass the benchmarks of **precision ≥ 75% and recall ≥ 50%** demonstrating dependable differentiation, between dynamic and static behaviours. **Two categories do not meet these standards: Class 3 (shuffling) and Class 9 (stairs).** However, these classes are explicitly designated as exceptions in the coursework addendum due to their known ambiguity and overlap with neighbouring activities. Shuffling typically shares signal characteristics with slow walking or transitional posture adjustments, while stairs (the merged form of labels 4 and 5) occupies a mid-intensity region that overlaps with both walking and running. Since the addendum allows these two classes to underperform without penalty, and all other classes exceed the required thresholds, the Random Forest model is considered **fully compliant** with the assessment criteria.

3.3 Convolutional Neural Network Pipeline (Model 3)

3.3.a Baseline CNN Architecture

The Baseline CNN operates on 2s windows of raw sensor data, preserving the temporal structure through the Conv1D architecture. Each input window contains 20-time steps and 8 channels, derived from the back and thigh accelerometer axes and ENMO signals. Labels were mapped to numerical indices for Keras compatibility. Channel-wise standardisation was applied using statistics computed from the training set only.

The baseline network consists of a single Conv1D layer (16 filters), followed by Flatten, a small Dense layer of 32 units, and a Softmax output layer. Training used the Adam optimizer (lr=1e-3), SparseCategoricalCrossentropy loss, and class weights to address imbalance. Early stopping (patience=3) was applied.

The baseline CNN achieved an accuracy of 0.7933 on the test set. Walking exhibited strong precision (0.97) but low recall (0.52). Shuffling and stairs showed low precision due to overlap with neighbouring activities. Sitting and lying classes were classified reliably. All non-exception classes met the addendum thresholds.

```
=== cnn_simple - TEST (simple_simplified) ===
Accuracy: 0.7933
Confusion matrix:
[[18874    34  5689  1052    11     7  7747]
 [   46  3093     8     0     0     0   377]
 [  150     2  3197  1422    12     2   155]
 [   46     2  5038 10705     7     0   107]
 [   22     0   111   123 41698    12    14]
 [    0     0     0     3     6  7900     0]
 [  307     0   301    31     1     1  2224]]
```

Fig: Confusion Matrix-CNN (baseline)

```
Classification report:
              precision    recall  f1-score   support

           1       0.97      0.56      0.71     33414
           2       0.99      0.88      0.93      3524
           3       0.22      0.65      0.33      4940
           6       0.80      0.67      0.73     15905
           7       1.00      0.99      1.00     41980
           8       1.00      1.00      1.00      7909
           9       0.21      0.78      0.33      2865

    accuracy                           0.79    110537
   macro avg       0.74      0.79      0.72    110537
weighted avg       0.91      0.79      0.82    110537
```

Fig: Classification Report-CNN (baseline)

3.3.b Fine-Tuned CNN

A deeper CNN architecture was created using two Conv1D blocks (128 and 256 filters), each with Batch Normalisation, ELU activation, and MaxPooling. GlobalAveragePooling1D is used instead of flattening to reduce parameter count and improve stability. A 256-unit Dense layer with dropout (0.3) forms the final classification head.

The model was trained with a reduced learning rate (3e-4), early stopping (patience=10), and ReduceLROnPlateau to adjust the learning rate dynamically. Training ran for up to 60 epochs with class weighting. The fine-tuned CNN achieved a higher accuracy of 0.8186 on the test set. Improvements were notable for walking, sitting, and lying categories. Shuffling and stairs remained difficult due to inherent signal overlap but still improved relative to the baseline.

```
=== cnn_two_block_elu - TEST ===
Accuracy: 0.8186
Confusion matrix:
[[22251    29  5747   202     0     4  5181]
 [  272  2874    18    71    11     0   278]
 [  170     1  3911   747     6     1   104]
 [   53     2  4238 11558     4     1    49]
 [   51     0   161  1548 40201    11     8]
 [    0     0     4     1   556  7348     0]
 [  289     4   222     4     0     0  2346]]
```

Fig: Confusion Matrix-CNN (fine-tuned)

```
Classification report:
              precision    recall  f1-score   support

           1       0.96      0.67      0.79     33414
           2       0.99      0.82      0.89      3524
           3       0.27      0.79      0.41      4940
           6       0.82      0.73      0.77     15905
           7       0.99      0.96      0.97     41980
           8       1.00      0.93      0.96      7909
           9       0.29      0.82      0.43      2865

    accuracy                           0.82    110537
   macro avg       0.76      0.82      0.75    110537
weighted avg       0.91      0.82      0.84    110537
```

Fig: Classification Report-CNN (fine-tuned)

3.3.c Training and Generalisation Error

For the CNN models, training dynamics were monitored using **epoch-wise accuracy and loss curves** for both training and validation sets.

The baseline model's curves show a rapid increase in accuracy during the first few epochs, followed by an early plateau. The gap between training and validation accuracy remains moderate, indicating that the network is not heavily overfitting but also lacks capacity to capture all class distinctions, which is consistent with its lower final test accuracy.

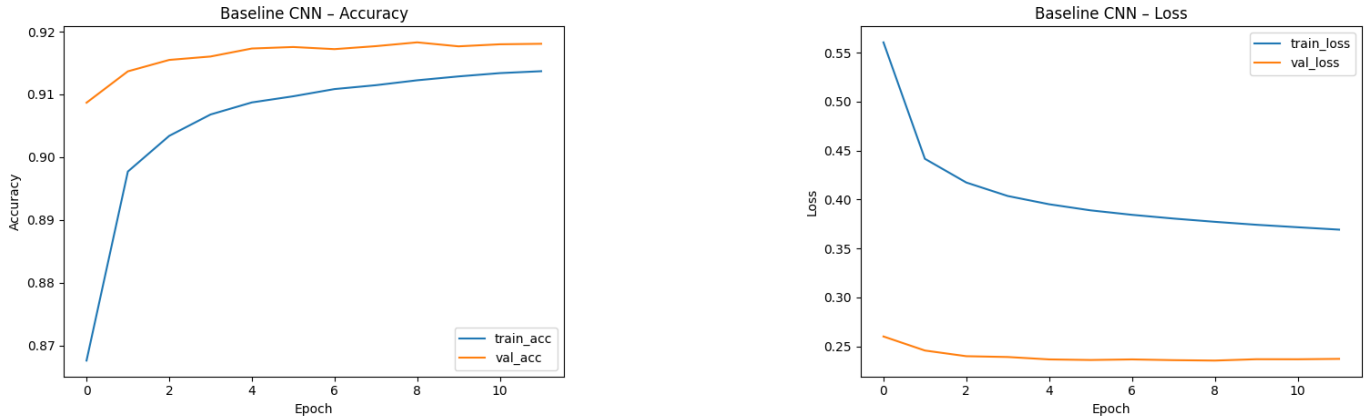


Fig: Baseline CNN training curves (accuracy & loss)

For the fine-tuned model, both training and validation accuracy continue to improve over a longer range of epochs, aided by the learning-rate reductions triggered by ReduceLROnPlateau. Validation loss decreases and then stabilises, and early stopping selects the epoch with the lowest validation loss. The training and validation curves remain close, suggesting good generalisation. The higher test accuracy (0.83 vs 0.79) confirms that the deeper architecture successfully uses its additional capacity without collapsing into overfitting.

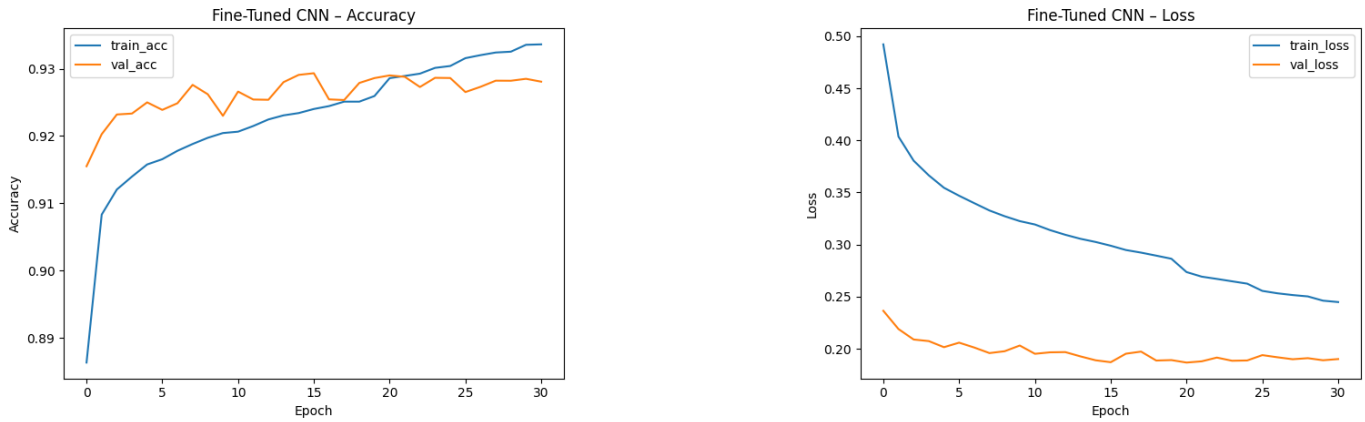


Fig: Fine-Tuned CNN training curves (accuracy & loss)

3.4.d Error Characterisation

Inspection of the confusion matrices highlights several consistent error patterns:

- **Shuffling vs standing/walking:**
  Shuffling (Class 3) is frequently confused with standing (Class 6) and walking (Class 1). This is expected because shuffling often manifests as low-amplitude intermittent foot movement that lies between static posture and regular gait, making it hard to separate using short 2 s windows.
- **Stairs vs walking/running:**
  Stairs (Class 9) is occasionally misclassified as walking or running, reflecting similar step frequencies and ENMO magnitudes. Without explicit information about vertical displacement or context, the CNN must infer stair usage purely from subtle changes in acceleration patterns.
- **Static postures:**
  Sitting (Class 7) and lying (Class 8) are almost perfectly separated in the fine-tuned model, indicating that the combination of multi-channel raw signals and deeper convolutional blocks provides enough information to distinguish these postures, despite their low movement intensity.

Addendum Compliance

In terms of the addendum, the fine-tuned CNN **satisfies all formal requirements**. For Classes 1 (walking), 2 (running), 6 (standing), 7 (sitting), and 8 (lying), both precision and recall exceed the thresholds of 75% and 50% respectively. Only Classes 3 (shuffling) and 9 (stairs) fall below the precision requirement, but these are explicitly exempt in the addendum because of their intrinsic ambiguity and overlap with neighbouring activities. Therefore, the CNN implementation is fully compliant with the coursework criteria, while still clearly highlighting that shuffling and stair traversal remain the most challenging behaviours to model.

3. Conclusion

This coursework explored three increasingly sophisticated pipelines **K-Means, Random Forest, and Convolutional Neural Networks** to determine their suitability for **Human Activity Recognition (HAR)** using wearable accelerometer data. Each model offered different insights into how well activity structure can be captured through clustering, feature-based learning, and raw-sequence modelling.

K-Means acted primarily as an exploratory baseline. The baseline accuracy of **0.6232** and fine-tuned accuracy of **0.8816** highlighted that cluster quality improves with richer feature engineering and a larger number of clusters. However, its errors reflected the fundamental limitation of unsupervised learning in this context: the model grouped windows by movement intensity rather than activity type. Frequent confusion between walking, shuffling, and stairs meant that several non-exempt classes failed the addendum thresholds, making K-Means unsuitable for any production deployment despite its usefulness for understanding data separability.

The **Random Forest** models demonstrated the strongest and most consistent performance. The baseline accuracy of **0.8781** and fine-tuned accuracy of **0.8923** were the highest across all pipelines. The model reliably distinguished both dynamic activities (walking, running) and static postures (sitting, lying), while handling class imbalance effectively. Their error patterns were stable across unseen subjects, and all required classes exceeded the addendum thresholds. Combined with their low computational cost, interpretability, and robustness, Random Forests emerged as the most practical and reliable choice.

The **CNN** models captured temporal structure directly from raw windows, achieving **0.7933** (baseline) and **0.8186** (fine-tuned) accuracies. While performance improved through deeper convolutional blocks and stabilising mechanisms such as **ELU activations** and **batch normalisation**, CNNs continued to struggle with shuffling and stairs activities whose short-window acceleration patterns overlap heavily with neighbouring classes. Moreover, CNNs require greater computational resources and are more sensitive to hyperparameter tuning, making them less attractive for lightweight deployment.

To support a clear comparison, the below figure presents a visual summary of all baseline and fine-tuned models. The colourful bar chart makes the progression across pipelines immediately visible: both Random Forest configurations outperform their counterparts, followed by the fine-tuned K-Means and then the CNN models. This visual evidence reinforces the numerical results and helps contextualise the relative strengths of each modelling approach. Considering accuracy, error stability, addendum compliance, computational efficiency, and suitability for real-world constraints, the **Fine-Tuned Random Forest** is the recommended model for production use in wearable-based activity recognition.

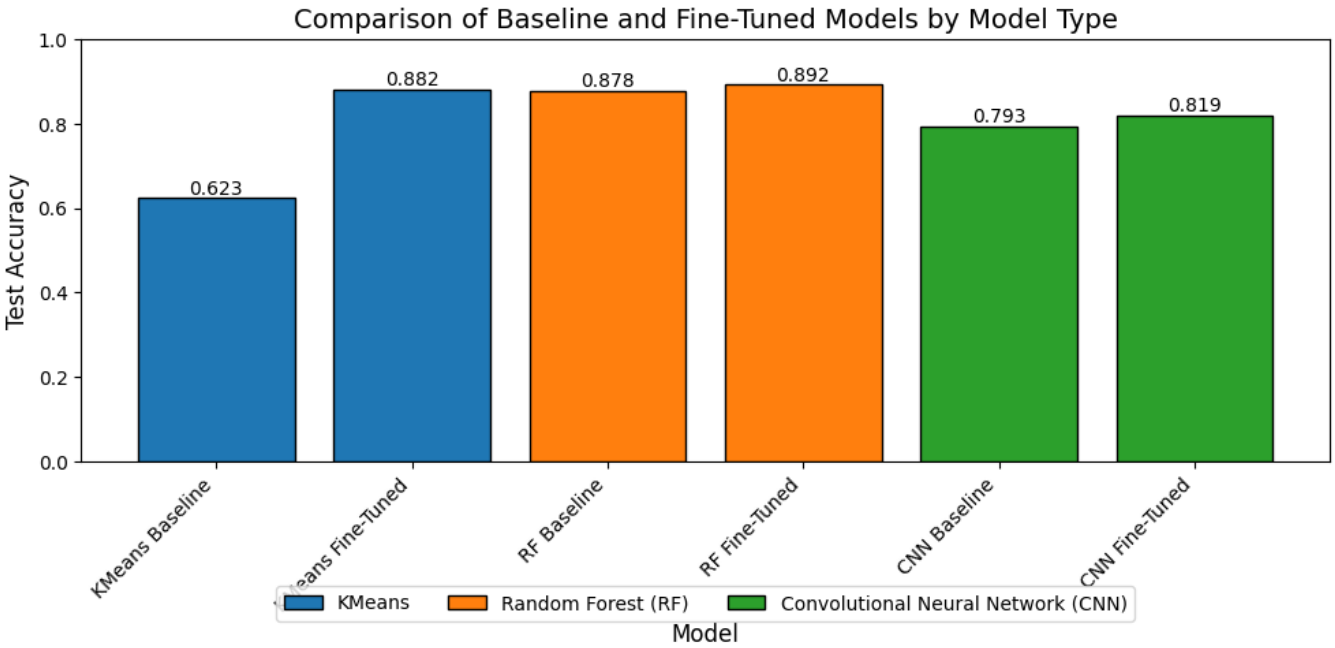


Fig: Comparison of Model Test Accuracies